Transfusion Policy Manual

Care and Management of Transfusion Reactions

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Clinical staff

Introduction

The purpose of this policy is to provide direction for recognising a suspected transfusion reaction and to ensure the safe and correct management of the patient. Transfusion reactions may be harmful, and potentially fatal, to patients, therefore it is important that the risk of such incidents are identified and the risk minimised. It is important that (suspected) transfusion reaction events are recognised promptly, managed appropriately, and reported as per SCGOPHCG and WA Department of Health (WA DoH) Haemovigilance reporting requirements.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☒
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

Policy Principles

Transfusion reactions may present with one, or multiple, of the following signs and symptoms:

• Temperature ≥ 38°C • Chills/ rigors/ flushing • Urticaria/ Hives/ Rash • Hyper/ hypotension • Tachycardia/ arrhythmia • Dyspnoea/tachypnoea/hypoxemia • Wheeze/ stridor/ pulmonary oedema	• Respiratory distress • Clinical Shock • Pain- Chest, Loin, Back, Infusion site • Unexplained bleeding e.g. haematuria • Nausea /Vomiting • Jaundice • Anxiety/ restlessness

The most significant reactions include acute and delayed haemolytic transfusion reactions, febrile non-haemolytic transfusion reaction, allergic, bacterial contamination of blood products, anaphylaxis, and Transfusion-Related Acute Lung Injury (TRALI). When the initial symptoms of a transfusion reaction occur, it may be difficult to ascertain which type of reaction has occurred.

Common transfusion adverse reactions and appropriate clinical actions can be found at appendix 1.

The transfusion medicine unit (TMU SCGH extn 34018)/ PathWest (OPH extn 78088) should be contacted if there are any concerns with issued blood products.

Procedure

If a transfusion reaction is suspected, immediate management must include the following:

1. **STOP** the transfusion- DO NOT disconnect
2. Check the patients' vital signs and assess if transfusion reaction is mild OR moderate to severe
3. For **MILD** transfusion reaction: that is patient may be experiencing fever ≥38°C AND >1°C from their baseline (but less than 39°C) +/- chills/rigors, or develop a localised/mild allergic reaction, but is otherwise haemodynamically stable
    a. Contact treating medical team
    b. Check that the patient details match on the blood pack **and** the patient
    c. vital signs every 15 minutes (or as directed by medical officer) until stable
    d. Administer supportive medications
    e. If symptoms are controlled, patient observations stable AND medical team in agreeance, transfusion can be recommenced at a slower rate with caution and close observation. Rate can be increased to prescribed rate if safe to do so (ensure transfusion time does not exceed 3 hours)
    f. If transfusion reaction symptoms increase or patient becomes unstable **STOP** the transfusion and follow MODERATE TO SEVERE management.
4. For **MODERATE TO SEVERE** transfusion reaction (haemodynamically unstable, fever ≥ 39°C, anaphylaxis, airway compromise, MET criteria)
    a. MET CALL if meets criteria/ Contact treating medical team
    b. Check that the patient details match on the blood pack **and** the patient
    c. Disconnect blood product (retain line and remaining blood product to send to TMU)
    d. Maintain IV access- flush with normal saline. Note: remove extension set prior to flushing to minimise further blood product entering patient
    e. Vital signs every 15 minutes until stable or as directed by medical order
    f. **SCGH** inform the PathWest Transfusion Medicine unit (TMU) on 6383 4018 or **OPH** PathWest 6457 8088
    g. Send to TMU (with the transfusion reaction report form & implicated unit and line) the following patient samples (date, time and sign) and request forms for the following tests:
        - 6mL EDTA (pink top) for G&S, crossmatch, DAT
        - Standard EDTA for FBC, reticulocyte and film

    - Lithium Heparin for UEC, LFT, LDH, Haptoglobin
    - Clotted sample for immunology HLA testing (only if allergic/anaphylactic reaction)
    - BNP (if respiratory distress)
5. Document transfusion reaction incident in the patient medical record
6. Complete a Transfusion Reaction and Adverse Incident Report (TRAIR)
7. The doctor may order blood cultures from the patient if febrile.

Possible Bacterial Contamination

If bacterial contamination suspected (Temperature ≥39°C or a change of ≥2°C from pre-transfusion value, sudden onset chills/rigours)

- Send the blood bag and attached IV administration set back to PathWest TMU via HSA
- DO NOT send spiked blood bags via the pneumatic chute system!
- Both the patient and the blood bag will require aerobic and anaerobic cultures as soon as possible.
- TMU/ blood bank will advise if additional blood tests are required for the patient

For quick reference guide to the treatment and management of the patients with suspected transfusion reaction, refer to The Transfusion Reaction Algorithm

Haemovigilance Reporting

Haemovigilance is the surveillance of the transfusion chain, from donor to recipient. It is intended to collect and investigate unexpected/undesirable events related to the transfusion chain and transfusion of blood products, to prevent their occurrence or recurrence.
All transfusion/suspected reactions MUST have a TRAIR form completed and forwarded to PathWest transfusion medicine.
The Transfusion Clinical Nurse Consultant is responsible for the clinical investigation and onward reporting of transfusion reaction/ haemovigilance events at SCGOPHCG.
The SCGOPHCG haemovigilance reporting flow chart can be located at appendix 2.

Further information related to haemovigilance can be found at Haemovigilance (health.wa.gov.au) and Australian Haemovigilance minimum data set | National Blood Authority

Compliance, monitoring and evaluation

Non- adherence to this policy may result in poor patient outcomes through inadequate or delayed management, treatment and reporting of suspected transfusion reactions.
The SCGOPHCG blood management committee is responsible for ensuring the effectiveness of this policy and monitors compliance through second monthly review of reported transfusion reactions at the Haemovigilance sub-committee meetings, review of clinical incidents and monitoring transfusion documentation compliance .

Related Documents

Standards, procedures, guidelines:

- ANZSBT Guidelines for the Administration of Blood Products
- WA Haemovigilance Reporting Guideline

Supporting documents

- Transfusion Reaction Algorithm
- Transfusion Reaction and Adverse Incident Report (TRAIR)

References

1. Australian & New Zealand Society of Blood Transfusion [ANZSBT] and Australian College of Nursing. Guidelines for the administration of blood products 3rd ed [Internet]. Sydney: ANZSBT and the Australian college of nursing; 2019 Oct [cited 2025 Jan 20]. Available from: https://anzsbt.org.au/wp-content/uploads/2020/03/ANZSBT-Administration-Guidelines-Revised-3rd-edition-Publication-Version-FINAL-20191002.pdf
2. Department of Health Western Australia (WA). WA Haemovigilance Reporting Guideline version 4 [Internet]. Perth: Department of Health WA; 2021 [cited 2025 Jan 20]. Available at https://ww2.health.wa.gov.au/~/media/Files/Corporate/general-documents/Haemovigilance/PDF/WA-Haemovigilance-Reporting-Guideline.pdf
3. National Blood Authority (NBA) Australia. Australian haemovigilance minimum data set version 1 [Internet]. Canberra: National Blood Authority Australia; 2015 Aug [cited 2025 Jan 20]. Available at https://www.blood.gov.au/australian-haemovigilance-minimum-data-set
4. Australian Red Cross Lifeblood. Management of suspected reactions [Internet]. Victoria: Lifeblood; 2021-2025 [cited 2025 Jan 20]. https://www.lifeblood.com.au/health-professionals/clinical-practice/adverse-events/management-of-suspected-reactions
5. New Zealand Blood Service (NZBS). Blood bank management of transfusion reactions due to possible bacterial contamination [Internet]. New Zealand: NZBS; 2021 Aug [cited 2022 Apr 12]. Available at: https://www.nzblood.co.nz/assets/Transfusion-Medicine/PDFs/136I090.pdf

Appendix 1.

Common transfusion adverse reactions and appropriate clinical actions			
Reaction	**Possible eitiology**	**Action**	**Investigation**
Fever ≥38°C and rise ≥1°C from baseline and/or chills/ rigors- NO other symptoms	**Febrile non-haemolytic transfusion reaction (FNHTR)**	STOP transfusion, do not disconnect. Medical review. Give anti-pyretic. Recommence cautiously if reaction subsides.	Nil if transfusion complete without incident. If cease transfusion, follow moderate/severe reaction management pathway
Fever ≥39°C or rise ≥2°C from baseline +/- other symptoms	**Bacterial contamination** (commonly also experience chills/rigors) or **acute haemolytic transfusion reaction** (AHTR- medical emergency, commonly also experience chills, rigors, dyspnoea, chest pain, flank pain, discomfort at infusion site, sense of dread, or abnormal bleeding, and may progress rapidly to shock.	STOP transfusion-do NOT restart. Check patient ID with blood product compatibility label. Full septic screen. Consider IV antibiotics. Maintain good urine output.	Sepsis work-up: blood cultures Incompatible blood work-up: Group, screen and DAT on pre and post-transfusion samples. Haemolysis work-up: FBC, LDH, bilirubin, haptoglobin, electrolytes, creatinine, urinalysis. DIC work-up: DIC may complicate a severe reaction – perform aPTT, PT, fibrinogen, D-Dimer (or FDP)
Rash or Urticaria (hives) no other symptoms, patient otherwise well	**Minor allergic reaction**	STOP transfusion- do NOT disconnect. Antihistamine. Recommence if reaction subsides.	Nil
symptomatic eg. tachycardia, periorbital oedema, tingling/swelling lips and or tongue	**Severe allergic reaction**	STOP transfusion- do NOT restart. Antihistamine +/- corticosteroid.	Nil
Acute dyspnoea, airway obstruction, hypotension	**Anaphylaxis** (consider IgA deficiency) (medical emergency)	STOP transfusion. Initiate basic life support. Notify Transfusion Medicine Unit (TMU) ASAP	Check haptoglobin, tryptase and IgA levels. Test for anti-IgA if IgA deficient.
Shortness of breath, dyspnoea, hypoxia	**Transfusion associated circulatory overload (TACO)**	STOP transfusion. Chest X-ray- assess patient for pulmonary oedema . Diuretics. Oxygen. Sit patient upright.	Check BNP/N-terminal pro-BNP. Elevated levels are more common in TACO.
	Transfusion-related acute lung injury (TRALI) (medical emergency)	STOP transfusion. Assess chest X-ray for infiltrates. Oxygen. Possible intubation, ventilation. **Notify TMU ASAP x34018**	HLA and HNA typing and antibodies. TRALI is a clinical diagnosis – investigations to exclude other reactions. Check BNP/N-terminal pro-BNP. Normal levels are more common in TRALI.
	Bacterial contamination or **acute haemolytic transfusion reaction (AHTR)** (medical emergency)	STOP transfusion-do NOT restart. Check patient ID with blood product compatibility label. Full septic screen. Consider IV antibiotics. Maintain good urine output.	Sepsis work-up: blood cultures Incompatible blood work-up: Group, screen and DAT on pre and post-transfusion samples. Haemolysis work-up: FBC, LDH, bilirubin, haptoglobin, electrolytes, creatinine, urinalysis. DIC work-up: DIC may complicate a severe reaction – perform aPTT, PT, fibrinogen, D-Dimer (or FDP)

Appendix 2.

SCGOPHCG Haemovigilance Reporting

Transfusion Reaction/Adverse Incident (Includes near misses)

- Notify medical team for patient assessment
- Notify PathWest Transfusion Medicine Unit (TMU) as required
- Document in patient's medical record
- Commence open disclosure

CLINICAL INCIDENT incl. NEAR MISS:

Person who identifies incident to complete online Datix CIMS

SAC 1 Incident /Sentinel Events
- must be reported to the Service Medical and Nurse Co-Directors within 24 hours of the event.

- Refer to NMHS Clinical Incident Management Policy

SAC 2/SAC 3 Incidents investigated at local level (i.e. ward/ department)

Review by Clinical Risk Management (CRM)
- notifies CNC Transfusion
- managed as per NMHS Clinical Incident Management Policy

SUSPECTED TRANSFUSION REACTION

Ward staff to complete Transfusion Reaction and Adverse Incident Report (TRAIR) form and send to PathWest transfusion lab

Transfusion lab sends TRAIR to CNC Transfusion for investigation

Transfusion CNC investigates TRAIR and presents report to the haemovigilance (HV) subcommittee for final outcome assessment

PathWest Transfusion scientists report serious (suspected) transfusion reactions, e.g. TRALI, bacterial contamination, to Lifeblood ASAP

TRAIRS related to **fractionated/ manufactured products** are reported to the TGA/ manufacturer by the CNC Transfusion

DoH WA HV program
- Transfusion CNC reports via the WA HV REDCap reporting tool

All HV events reported to Blood Management Committee
- Discussion and review of incidents
- Assess recommendations

Authorisation & Version Control

Policy Sponsor	Executive Sponsor Blood Management Committee				
Policy Contact	Transfusion Clinical Nurse Consultant				
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Standards Applicable	NSQHS Standards (Version 2):				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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